

# DATA QUALITY REPORT

## IndiaKart E-Commerce Analytics

The purpose of this report is to assess the quality and analytical readiness of the IndiaKart datasets before SQL analysis, Exploratory Data Analysis (EDA), KPI calculations, and Power BI dashboard development.

The assessment covers **8 interconnected datasets with approximately 216,200 records**: Customers, Orders, Order Items, Products, Payments, Returns, Inventory, and Suppliers. IndiaKart\_Project\_PRD.

### Executive Summary

Overall, the IndiaKart data demonstrates **good structural quality and is suitable for downstream analysis**, with a few issues requiring attention.

No duplicate primary identifiers were detected in the tested datasets. The major data-quality issue was found in the **Orders** table, where **17,501 of 50,000 records (35.0%) have a null delivered\_date**. These nulls may be valid for orders that are still Processing, Shipped, or Cancelled and therefore require business-context validation.

The assessment also identified **5,989 records where order\_date is later than delivered\_date**, which violates the expected order lifecycle. In addition, **312 orders with final\_amount greater than ₹500,000** were identified as potential outliers and flagged for review.

The Payments dataset contains **1,748 failed transactions out of 50,000 payments**, resulting in a **3.5% payment failure rate**. Furthermore, **457 customers have zero orders**, while **2 products are classified as Out of Stock**. These represent valid business conditions rather than automatic data-quality errors.

Several date-related columns were also standardized from object/string values into appropriate date formats to support reliable time-based analysis.

### 2. Data Quality Summary

| Dataset     | Records | Key Finding                                              | Assessment            |
|-------------|---------|----------------------------------------------------------|-----------------------|
| Customers   | 10,000  | 457 customers have zero orders                           | Clean                 |
| Orders      | 50,000  | 17,501 null delivery dates; 5,989 invalid date sequences | Requires Attention    |
| Order Items | 100,000 | No duplicates; Order links validated                     | Clean                 |
| Products    | 1,000   | IDs unique; dates standardized                           | Clean                 |
| Payments    | 50,000  | 1,748 failed payments                                    | Business Condition    |
| Returns     | 10,000  | Return-to-Order links validated                          | Clean                 |
| Inventory   | 1,000   | 2 Out-of-Stock products                                  | Operational Condition |
| Suppliers   | 200     | No nulls; IDs unique                                     | Clean                 |

### Key Findings

#### Orders

No duplicate order\_id values were detected. However, **17,501 Orders (35.0%) have missing delivered\_date values**. These should be evaluated against order status before being classified as errors.

A total of **5,989 records have order\_date > delivered\_date**, indicating an invalid chronological relationship. These records should be investigated before calculating delivery-time or logistics KPIs.

Additionally, **312 Orders have a final\_amount above ₹5 lakh** and were flagged as potential outliers, as required by the project quality checks. IndiaKart\_Project\_PRD

## Customers

No duplicate customer identifiers or null values were identified. **457 customers have zero orders**, which is considered acceptable because registered customers may not yet have completed a purchase. The PRD specifically identifies this as a condition to note rather than an error. IndiaKart\_Project\_PRD

## Payments

Out of **50,000 payment transactions, 48,252 were successful and 1,748 failed**, resulting in a **3.5% payment failure rate**. Failed payments are valid transactional events and should be retained for subsequent business analysis.

## Inventory

The assessment identified **2 Out-of-Stock products**. These records represent an operational condition and should be retained for inventory and stock-availability analysis.

## Order Items & Returns

No duplicate identifiers were detected in the tested data. Referential-integrity checks confirmed that tested Order Item and Return order\_id values correspond to valid records in the Orders dataset.

### DATA QUALITY REPORT — DETAILED ASSESSMENT

#### 4. Validation Results

| Validation                      | Finding          | Result                        |
|---------------------------------|------------------|-------------------------------|
| Customer ID uniqueness          | 0 duplicates     | <b>Passed</b>                 |
| Customers with zero orders      | 457              | <b>Valid Condition</b>        |
| Order ID uniqueness             | 0 duplicates     | <b>Passed</b>                 |
| Missing delivered_date          | 17,501           | <b>Requires Review</b>        |
| order_date > delivered_date     | 5,989            | <b>Requires Investigation</b> |
| Orders above ₹5 lakh            | 312              | <b>Flagged for Review</b>     |
| Order Item → Order relationship | Valid references | <b>Passed</b>                 |
| Failed payments                 | 1,748 / 50,000   | <b>3.5% Failure Rate</b>      |
| Return → Order relationship     | Valid references | <b>Passed</b>                 |
| Out-of-Stock products           | 2                | <b>Valid Condition</b>        |

**Issues Requiring Attention**

## Invalid Order & Delivery Dates

The most important data-quality issue is the **5,989 records where the order date occurs after the delivery date**. This violates the expected business sequence:

**Order Placed → Processing → Shipped → Delivered**

These records could distort delivery-duration and logistics-performance calculations.

**Action:** Investigate the affected records against the source data before using them in delivery-related analysis.

**Status:** Requires Investigation

### Missing Delivery Dates

A total of **17,501 Orders have a null delivered\_date**. These values should not automatically be treated as missing-data errors because Processing, Shipped, or Cancelled orders may legitimately have no delivery date.

**Action:** Validate missing delivery dates against the corresponding order status and retain legitimate null values.

**Status: Requires Business Validation**

### High-Value Orders

**312 Orders exceed ₹500,000 in final\_amount** and were flagged as potential outliers.

**Action:** Review these transactions for bulk purchases, premium products, unusual quantities, or pricing anomalies. Valid high-value transactions should be retained.

**Status: Flagged for Review**

### Date Data Types

Several date-related columns were initially stored as object/string values.

**Action Taken:** Relevant date columns were standardized using Pandas to support accurate sorting, filtering, date comparisons, time-series analysis, and reporting.

**Status: Resolved**